Skills Summary

Sorting Shapes by What Counts

Use this reference while completing your worksheet or when studying.

Sort by the number of sides

Rule: Count the straight sides of each shape, then group the equal counts: 3 sides → triangles, 4 sides → squares and rectangles, 6 sides → hexagons.

Example: A set has a triangle, a square, a tilted triangle, and a rectangle. How many shapes go in the 3-sided group?
Step 1. Count sides one shape at a time — tilt does not change a count.
Step 2. Triangle: 3. Square: 4. Tilted triangle: 3. Rectangle: 4. Two shapes have 3 sides.
⇒ the 3-sided group holds **2** shapes

Try it: A set has 8 shapes, and 5 of them have four sides. How many shapes are NOT in the 4-sided group?

check: **3**

Sort by the number of corners

Rule: Corners work just like sides: count where two sides meet. A shape with 3 sides always has 3 corners. After sorting, you can compare group sizes by subtracting.

Example: After a corner sort, the 3-corner group has 5 shapes and the 4-corner group has 3 shapes. How many more shapes are in the 3-corner group?
Step 1. Comparing two group sizes means finding their difference — subtract the counts.
Step 2. $5 - 3 = 2$.
⇒ the 3-corner group has **2** more shapes

Try it: One group has 7 shapes and another has 4. How many more shapes are in the bigger group?

check: **3**

Defining attributes, not looks

Rule: DEFINING attributes never change: number of sides, number of corners, closed or open. Color, size, and tilt are just looks — they can change without changing the shape's name.

Example: A big gray triangle and a tiny white triangle: how many sides does each one have?
Step 1. Ignore the looks (big, small, gray, white) and count what defines the shape.
Step 2. Each one has exactly 3 straight sides — so both are triangles, same group.
⇒ each has **3** sides

Try it: A tilted square and an upright square: how many equal sides does each have?

check: 4 — same group!

(!) Watch out: "It looks different" is not a sorting reason. If you cannot COUNT it, do not sort by it.